

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – REAL NUMBERS (1.1) – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Think back to Sister Wanjiku at the Mathare North Health Centre in Nairobi. Her calculator gave a neat answer for the average temperature, $(36.8 + 37.0 + 37.5) \div 3$, but a never-ending decimal for the diagonal of the 1 m square waiting-room tile. In the space under each prompt, write your own explanation using evidence from our lessons. Use correct vocabulary (rational, irrational, p/q , terminating, repeating, reciprocal). Show your working for any calculation. Give reasons for your claims, not just answers. Write in full sentences.

Part 1: What kind of numbers did Sister Wanjiku record?

Sister Wanjiku recorded body temperatures of 36.8 °C, 37.0 °C and 37.5 °C. She also recorded blood pressure readings of 120/80 and 135/90, and shoe sizes of 40, 41 and 43. (a) State what it means for a number to be rational. (b) Write each of the three temperatures as a fraction p/q with p and q integers and $q \neq 0$. (c) Explain whether the shoe sizes also belong to the set of rational numbers, and say which other number sets they belong to.

(a) A rational number is any number that can be written as p/q , where p and q are integers and $q \neq 0$. Its decimal either terminates or repeats. (b) $36.8 = 368/10 = 184/5$. $37.0 = 37/1$. $37.5 = 375/10 = 75/2$. Each is a ratio of two integers, so each is rational. (c) Yes. Shoe sizes 40, 41 and 43 are whole numbers, so they are natural numbers ($\mathbb{N}$) and integers ($\mathbb{Z}$). Every integer can be written as itself over 1 (for example $40 = 40/1$), so they are also rational ($\mathbb{Q}$) and real ($\mathbb{R}$). This follows the nested structure $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.

Part 2: Why did the average temperature 'end neatly'?

Sister Wanjiku calculated $(36.8 + 37.0 + 37.5) \div 3$. (a) Work out the answer, showing each step. (b) Write your answer as a fraction p/q . (c) Using what you learned about operations on rational numbers, explain what

(a) $36.8 + 37.0 + 37.5 = 111.3$. Then $111.3 \div 3 = 37.1$. (b) $37.1 = 371/10$, which is a ratio of two integers. (c) All three temperatures are rational. Rational numbers are closed under addition: if $a = p/q$ and $b = m/n$, then $a + b = (pn + mq)/(qn)$, which is still a ratio of integers. They are also closed under division by a non-zero integer such as 3. So adding three rational numbers and dividing by 3 must give a rational number. A rational number has a decimal that terminates or repeats, so the calculator display was never going to run on forever without a pattern. Here it terminates at 37.1.

you could say about the type of number the answer would be before using the calculator, and give your reasons.	
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Part 3: Why does the tile diagonal go on forever?

The waiting-room floor tile is a square with side 1 m. (a) Use the Pythagorean theorem to write an expression for the diagonal length. (b) Explain what is different about this number compared with Sister Wanjiku's average temperature. (c) Sister Wanjiku's calculator shows 1.41421356... A classmate says, 'The calculator display stops, so the decimal must terminate.' Explain why this reasoning is not safe. (d) Name one other irrational number we met in this topic, and explain how 22/7 differs from π.	(a) Diagonal = $\sqrt{1^2 + 1^2} = \sqrt{2}$ metres. (b) The average temperature came from adding and dividing rational numbers, so it is rational (371/10). The number $\sqrt{2}$ comes from taking the square root of 2, which is not a perfect square. No fraction p/q with p and q integers and $q \neq 0$ satisfies $(p/q)^2 = 2$ exactly, so $\sqrt{2}$ is irrational. Its decimal neither terminates nor repeats. The difference is in the nature of the number, not in the calculator. (c) A calculator screen can show only a limited number of digits, so it cuts the decimal off. A display that ends is a truncated display, not proof that the decimal terminates. Responsible calculator use means knowing whether the number is rational or irrational before treating the screen value as exact. 1.41421356 is only an approximation of $\sqrt{2}$. (d) π is another irrational number. 22/7 is rational because it is a ratio of two integers, and its decimal repeats (3.142857 repeating). π is irrational and its decimal does not repeat. The two values are close, but 22/7 is not equal to π.
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Part 4: Reciprocals

(a) Define the reciprocal of a non-zero real number x. (b) Find the reciprocal of 37.1 as a fraction. (c) Explain why the reciprocal of $\sqrt{2}$ is irrational. (d) Which number has no reciprocal, and why?	(a) The reciprocal of a non-zero real number x is $1/x$, and $x \times (1/x) = 1$. (b) $37.1 = 371/10$, so its reciprocal is $10/371$. This is a ratio of integers, so it is rational. (c) The reciprocal of $\sqrt{2}$ is $1/\sqrt{2}$. Suppose $1/\sqrt{2}$ were rational, say p/q with p and q non-zero integers. Then $\sqrt{2} = q/p$, which would make $\sqrt{2}$ rational. But we showed that $\sqrt{2}$ is irrational. This is a contradiction, so $1/\sqrt{2}$ must be irrational. (d) Zero has no reciprocal. No number multiplied by 0 gives 1, and $1/0$ is not defined.
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Part 5: Answering the driving question

Driving question: Why do some numbers from our everyday Kenyan lives 'end neatly' while others go on forever, and how does knowing the difference	Numbers that can be written as p/q, with p and q integers and $q \neq 0$, are rational, and their decimals either terminate or repeat. Sister Wanjiku's temperatures are rational, and rational numbers are closed under addition and division, so her average, $(36.8 + 37.0 + 37.5) \div 3 = 37.1 = 371/10$, ends neatly. The tile diagonal $\sqrt{2}$ goes on forever because 2 is not a perfect square and no fraction p/q equals $\sqrt{2}$ exactly, so it is irrational. Knowing the difference tells us when a calculator answer is exact and when it is only an approximation. For example, Sister Wanjiku can record her average of 37.1 °C as exact, but a builder cutting a 1 m square
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help us make better decisions? Write a clear answer of 5–7 sentences that uses Sister Wanjiku's data (the average temperature and the tile diagonal). Include one example from health, construction or commerce showing why knowing whether a number is rational or irrational matters.	tile diagonally must use an approximation such as 1.414 m and allow for a small error. Knowing which type of number we have helps us decide when to round.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classification of numbers (rational vs irrational)	Defines rational and irrational numbers accurately, writes every health-centre number as p/q correctly, and places numbers in $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ with correct reasons.	Defines the terms mostly correctly and writes most numbers as p/q, with minor errors in set placement or reasons.	Definitions are incomplete or incorrect, or learner cannot write numbers as p/q or place them in number sets.
Use of closure and operations on rational numbers	Calculates 37.1 correctly with clear steps and uses closure under addition and division to justify why the result is rational.	Calculates the average correctly and mentions closure, but the justification is incomplete.	Calculation contains errors or the learner gives no reason for why the result ends neatly.
Explaining irrational numbers and calculator use	Explains why $\sqrt{2}$ is irrational, why a truncated display does not prove a decimal terminates, and distinguishes 22/7 from π accurately.	Identifies $\sqrt{2}$ as irrational and notes the calculator limitation, but the reasoning or the 22/7 versus π distinction is partly unclear.	Treats the calculator display as exact or cannot explain why $\sqrt{2}$ differs from the average temperature.
Reciprocals and mathematical reasoning	Defines reciprocals, finds 10/371 correctly, gives a correct proof by contradiction for $1/\sqrt{2}$, and states that zero has no reciprocal.	Finds correct reciprocals and states $1/\sqrt{2}$ is irrational with a partial argument.	Reciprocal definition or calculations are incorrect and no reasoning is offered.
Final explanation linked to Kenyan context and decisions	Gives a coherent answer to the driving question using Sister Wanjiku's data and a realistic decision example.	Answers the driving question using the data, with a weak or general decision example.	Answer is vague, not linked to the data, or gives no decision example.

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